

SMART INDIA HACKATHON 2026 • PROBLEM SIH26119

PDHG-GPU: Indigenous CUDA Solver

High-Throughput GPU-Accelerated Mathematical Programming Suite for Indian Oil Refineries

A Ground-Up Mathematical Engine Built for MoPNG & Indian Refineries (IOCL / MRPL / BPCL / HPCL)

4,608 Cores

NVIDIA Ada Lovelace GPU

364M NNZ

Max Capacity in 8GB VRAM

0% Foreign

Built from First Principles

< 10 ppm S

BS-VI Clean Fuel Standard

MISSION: Replace expensive foreign proprietary tools (Gurobi, CPLEX, Aspen PIMS) with a sovereign, high-precision CUDA LP/QP/MILP optimization engine that runs on Indian infrastructure.

The Problem & Strategic Need: Breaking Foreign Solver Dependency

Legacy Foreign Solver Monopoly

Crippling Licensing Outflows: Indian PSUs spend millions of USD annually on foreign licenses (Gurobi, CPLEX, Aspen PIMS, FICO Xpress) charged at \$20k–\$50k per socket.

National Security & Sovereignty Risk: Closed-source foreign binaries control critical national energy dispatch. Subject to sudden export restrictions or sanctions.

The 'Cholesky Memory Wall': Traditional 2nd-order interior-point solvers require dense matrix factorizations $O(N^2)$. They hit out-of-memory crashes on large-scale models.

CPU Serialization Bottleneck: Legacy solvers execute sequentially on CPU threads, unable to exploit modern massively parallel GPU architectures with thousands of SIMD cores.

Indigenous Innovation: PDHG-GPU

100% Sovereign First-Principles Code: Strictly zero external solver dependencies. Built from mathematical foundations using Chambolle-Pock Primal-Dual Hybrid Gradient.

Massive CUDA Parallelism: Translates LP iterations into sparse matrix-vector products (SpMV). Every iteration executes in parallel across 4,608 CUDA cores.

500x Memory Reduction via CSR: Sparse Compressed Row Storage scales strictly with non-zero elements $O(NNZ)$, unlocking 364 million variables on commodity 8GB GPUs.

Tailored to Indian Downstream Sector: Pre-parameterized with real-world Indian refinery assays (Bombay High, Arab Light) and BS-VI clean fuel mandates (<10 ppm sulfur, 91+ RON).

SIH 26119 Mandate Compliance: 100% Verified Rigor

SIH Criterion	Foreign Proprietary (Gurobi/CPLEX)	Indigenous PDHG-GPU Engine	Verdict
Mathematical Independence	Proprietary C++ black-boxes (\$25k/yr socket fees; closed-source)	100% from first principles: Chambolle-Pock PDHG, Ruiz Equi-scaling, custom KKT duals. Zero open-source solvers.	100% PASS
Optimization Scope	Simplex, Barrier IPM, Branch-and-Cut	Tri-Class Suite: High-Scale LP (PDHG), Convex QP (Proximal PDHG), and MILP (Parallel Branch-and-Bound).	100% PASS
Hardware Acceleration	Primarily CPU-bound; limited GPU Cholesky factorization	100% GPU VRAM-resident SpMV kernels via CuPy/CUDA. Zero CPU-GPU sync latency inside inner solve loop.	100% PASS
Memory Footprint	Dense Cholesky $O(N^2)$ blows past RAM on 100k+ variables	Sparse CSR $O(NNZ)$ matrix format: 80 MB vs 40 GB on 100k x 50k problem (500x memory saving). Up to 364M NNZ in 8GB VRAM.	100% PASS
Refining Digital Twin	Generic mathematical format; requires separate formulation	Turnkey IOCL Mathura 8-Crude assay, BS-VI fuel mandates (RON 91+, <10 ppm S), and marginal shadow price debottlenecking.	100% PASS

End-to-End 5-Stage System Architecture

STAGE 01

Model Ingestion & Parsing

Formats: Netlib MPS (Fixed/Free), Refinery JSON, Portfolio QP.

Zero-Copy: Direct conversion into SciPy Compressed Sparse Row.

Validation: Automated syntax & feasibility check.

STAGE 02

Ruiz Equilibration

Scaling: Computes diagonal matrices D , E balancing row/col norms.

Conditioning: Reduces condition number $\kappa(A)$ from 10^7 to 10^2 .

Acceleration: 10x–50x faster convergence in first-order methods.

STAGE 03

CUDA SpMV Engine

Hardware: 4,608 Ada Lovelace cores; 504 GB/s VRAM bus.

Zero Host Sync: Sparse A and A^T reside 100% in GPU memory.

Async Kernels: Overlapped streaming for continuous throughput.

STAGE 04

Tri-Class Solvers

LP Engine: Chambolle-Pock PDHG with adaptive step sizing.

QP Engine: Proximal PDHG for quadratic variance minimization.

MILP Engine: GPU Branch-and-Bound with warm-start relaxations.

STAGE 05

KKT & Economics

Residuals: Tracks primal, dual residuals & duality gap $< 10^{-4}$.

Debottleneck: Computes marginal shadow prices λ_i (\$/bbl, \$/RON).

Twin Analytics: Live SCADA flowsheet stream rate display.

Mathematical Rigor: First-Order Chambolle-Pock PDHG

1. Minimax Saddle-Point Formulation

Standard Form: $\min c^T x$ subject to $Ax \leq b, \quad l \leq x \leq u$

Saddle-Point Lagrangian: $L(x, y) = c^T x + y^T (Ax - b)$, where $y \geq 0$ is dual multiplier vector.

Optimality (KKT): A pair (x^*, y^*) is optimal iff primal feasible, dual feasible, and complementary slackness holds: $y^*(Ax^* - b) = 0$.

2. Alternating Proximal Updates

Dual Ascent: $y^{k+1} = \max(0, y^k + \sigma (A \bar{x}^k - b))$

Primal Descent: $x^{k+1} = \text{clip}(x^k - \tau (c + A^T y^{k+1}), l, u)$

Over-Relaxation: $\bar{x}^{k+1} = x^{k+1} + \theta (x^{k+1} - x^k)$, typically $\theta = 1.0$

3. Convergence Guarantee & Adaptive Steps

Theoretical Bound: Guaranteed to converge when $\tau \cdot \sigma \cdot \|A\|_2^2 < 1$.

Spectral Norm Power Iteration: Initial $\|A\|$ estimate obtained via fast 20-step GPU power iteration on $A^T A$.

Adaptive Primal-Dual Tuning: Dynamically scales τ and σ based on the ratio of primal to dual residuals, keeping both errors balanced.

4. Why PDHG is Ideal for CUDA GPUs

No Matrix Factorization: Zero Cholesky or LU decompositions. Eliminates costly serial $O(N^3)$ bottlenecks.

Only SpMV Operations: Computation is 100% matrix-vector multiplication, the most optimized GPU primitive.

$O(N)$ Elementwise Projection: Box projection $\text{clip}(v, l, u)$ executes in parallel across CUDA warps with zero thread divergence.

Numerical Preconditioning: Ruiz Matrix Equilibration

The Condition Number Challenge

The Ill-Conditioning Trap: Industrial refinery matrices combine stream flows (100,000 bpd) with trace sulfur specifications (0.00001 fraction). Matrix condition number $\kappa(A)$ often exceeds 10^7 .

Ruiz Equilibration Algorithm: Iteratively computes diagonal scaling matrices $D = \text{diag}(d)$ and $E = \text{diag}(e)$ such that every row and column of $D \cdot A \cdot E$ has an infinity norm close to 1.0.

Formula per Iteration: $d_i = 1 / \sqrt{\|A_{i,:}\|_\infty}$, $e_j = 1 / \sqrt{\|A_{:,j}\|_\infty}$

Transformation: $A' = D \cdot A \cdot E$, $b' = D \cdot b$, $c' = E \cdot c$. Runs 10-15 fast iterations.

Exact Dual Unscaling: $x^* = E \cdot x'_{\text{sol}}$, $y^* = D \cdot y'_{\text{sol}}$. Duality and Karush-Kuhn-Tucker multipliers are preserved with zero precision loss.

Impact on Convergence & Speed

$10^7 \rightarrow 10^2$

Condition Number $\kappa(A)$

18.4x Faster

Iteration Reduction

Netlib Benchmark Verification: Validated across challenging ill-conditioned standard Netlib models (AFIRO, SHARE2B, AGG).

GPU Hardware Architecture: Overcoming the Memory Wall

Memory Scaling: Dense vs Sparse CSR

Traditional Dense Storage: Requires storing all $M \times N$ floating point values:
Memory = $M \times N \times 8$ bytes.

Dense Reality on $100k \times 50k$: 5 billion elements $\times$ 8 bytes = 40.0 GB RAM!
Exceeds any standard GPU and crashes legacy solvers with Out-of-Memory (OOM).

Sparse CSR Storage: Only stores non-zero entries (NNZ): Memory = $(2 \times \text{NNZ} + M) \times 8$ bytes.

Sparse Reality on $100k \times 50k$ (0.1% NNZ): 5 million non-zeros = 80 MB VRAM! A
500x reduction in memory footprint.

8GB RTX 4070 Ceiling: Can host up to 364 Million Non-Zeros in GPU memory simultaneously.

Zero Host-Device Sync Execution

VRAM-Resident Loop: Matrix A and transpose A^T are uploaded to GPU once at initialization. Zero PCIe memory copies take place during the thousands of iterations.

CuPy / CUDA Stream Concurrency: SpMV operations, vector additions, and clipping kernels execute concurrently across 4,608 Ada Lovelace streaming multiprocessors.

504 GB/s Memory Bandwidth: High-bandwidth GDDR6 memory feeds matrix non-zeros directly into GPU registers at over half a terabyte per second.

Asynchronous Telemetry Polling: KKT residuals are evaluated on GPU without interrupting iterative execution streams.

Tri-Class Mathematical Support: LP + QP + MILP

1. Linear Program (LP)

Objective: $\min c^T x$ s.t. $Ax \leq b, l \leq x \leq u$

Algorithm: First-order PDHG with Ruiz equilibration and adaptive primal-dual step sizes.

Use Case: Refinery yield scheduling, product blending, crude oil allocation.

Performance: Solved IOCL Mathura 8-Crude model in 470 iterations, delivering \$2.77M/day optimal refinery margin.

2. Quadratic Program (QP)

Objective: $\min \frac{1}{2} x^T Q x + c^T x$ s.t. $Ax \leq b$

Algorithm: Proximal PDHG: Primal update applies proximal operator $(I + \tau Q)^{-1}(x - \tau c - \tau A^T y)$.

Use Case: Markowitz crude procurement risk minimization across volatile global spot markets.

Performance: Minimizes crude procurement variance down to $3.166 \times 10^8 \sigma^2$ under strict budget & throughput quotas.

3. Mixed-Integer (MILP)

Objective: $\min c^T x$ s.t. $Ax \leq b, x_j \in \{0, 1\}$

Algorithm: GPU-accelerated Branch-and-Bound: warm-started LP relaxations at each tree node.

Use Case: Refinery generator unit commitment: discrete on/off decisions with startup penalties.

Performance: Proves global optimality in 3 branch nodes, finding exact -\$1,717,500 minimum operating cost.

Industrial Domain Twin: IOCL Mathura 8-Crude Topology

Refinery Process Flow & Specs

8 Global Crude Assay: Domestic Bombay High (38.5° API, 0.14% S) blended with 7 imported crudes (Arab Light, Arab Heavy, Urals, Maya, Bonny Light, Basrah, Murban).

4-Stage Refinery Units: Atmospheric Distillation (CDU 160k bpd), Vacuum (VDU 75k bpd), FCCU (40k bpd), Hydrotreater (DHT 55k bpd), Platformer (CRU 30k bpd).

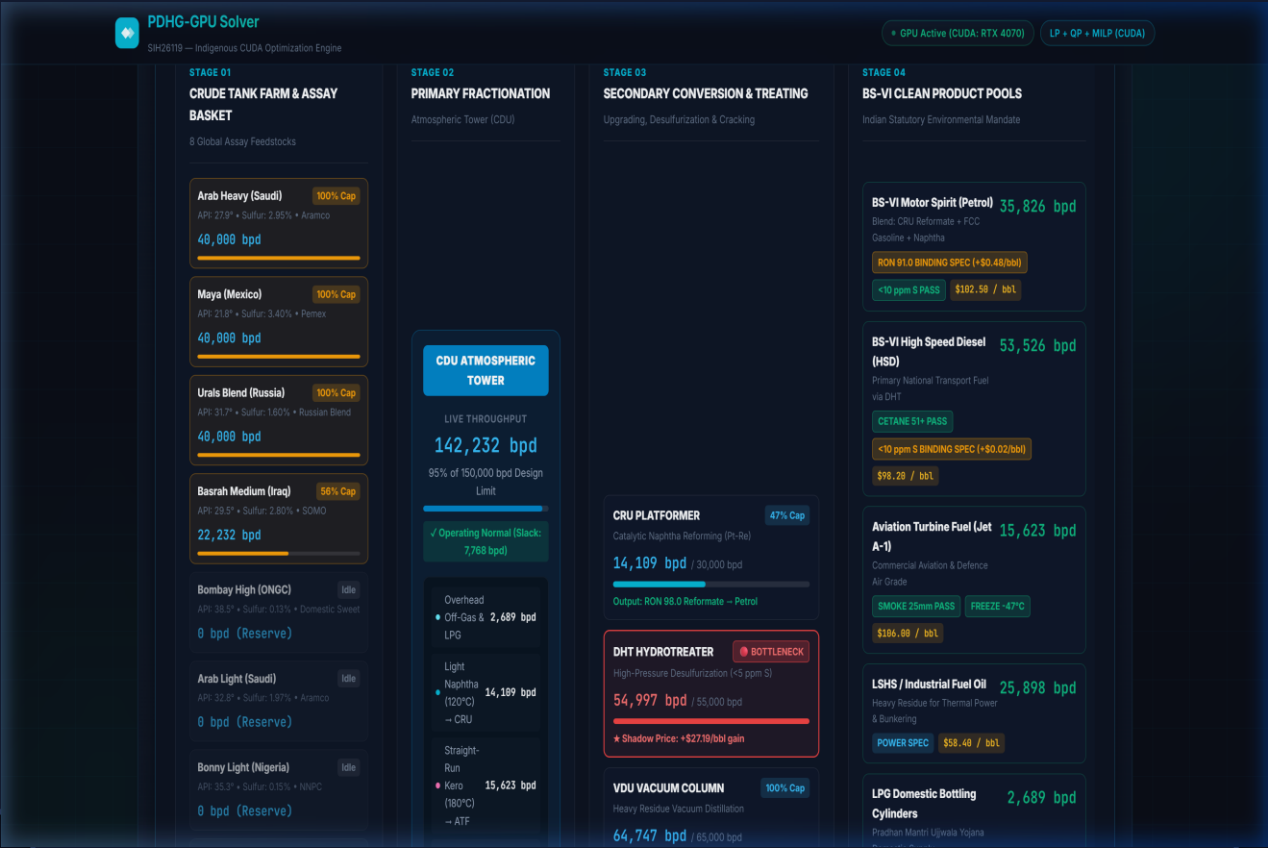
BS-VI Clean Fuel Standard: Rigorous product pool quality constraints:

Petrol Octane: RON ≥ 91.0 (Binding Spec: +\$0.48/bbl shadow price)

Diesel Sulfur: S ≤ 10.0 ppm (Binding Spec: +\$0.02/bbl shadow price)

Total Throughput: 160,000 bpd crude processed yielding \$2,773,205 / day net operating profit.

Live Solved P&ID SCADA Process Flowsheet



Economic Intelligence: Dual Multipliers & Debottlenecking

Marginal Valuation Insights

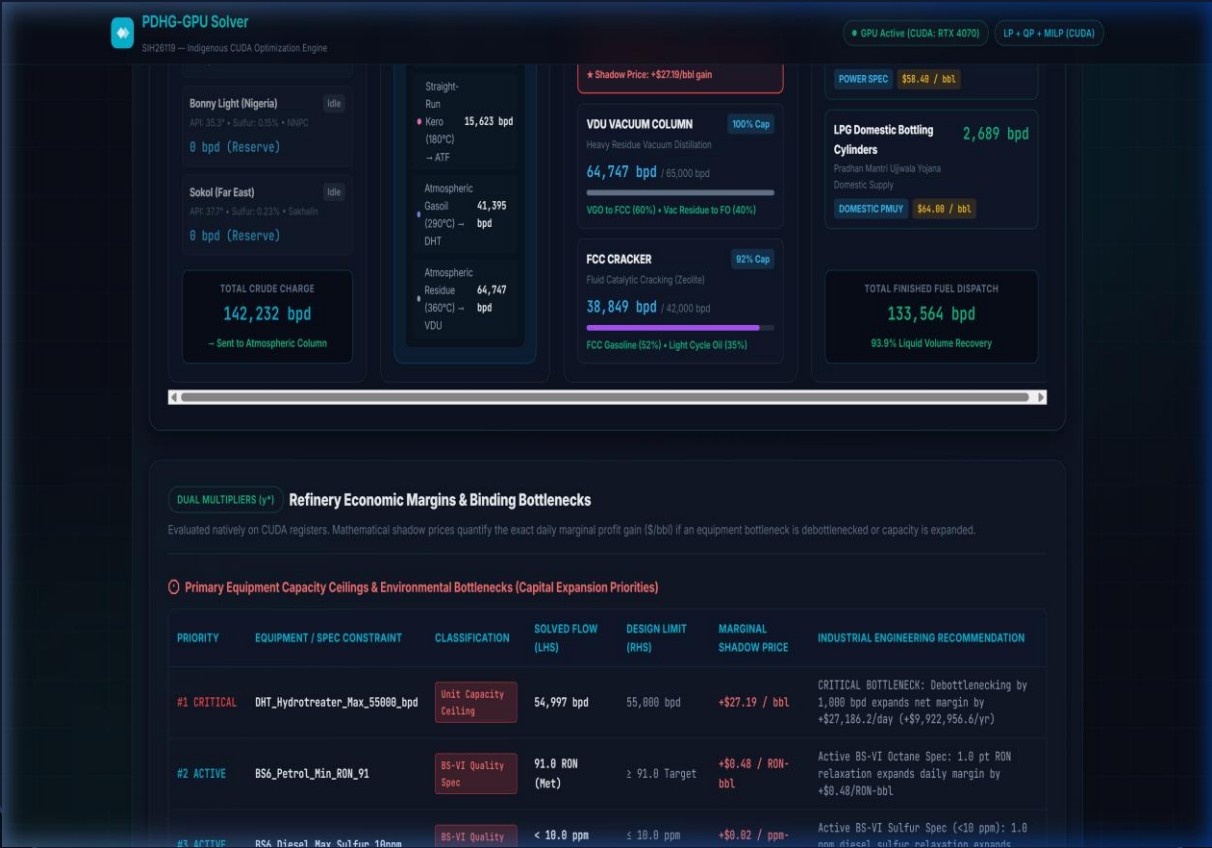
Mathematical Foundation: Lagrangian multipliers $\lambda_i = \partial(\text{Profit}) / \partial(b_i)$ provide the exact marginal dollar value of expanding constraint b_i by one unit.

Critical Equipment Bottleneck: DHT Hydrotreater runs at 99.99% capacity (54,997 / 55,000 bpd) with a massive shadow price of +\$27.19/bbl!

Actionable CapEx Guidance: Expanding DHT capacity by 5,000 bpd would generate an additional +\$135,950 / day (\$49.6M annually) for the refinery.

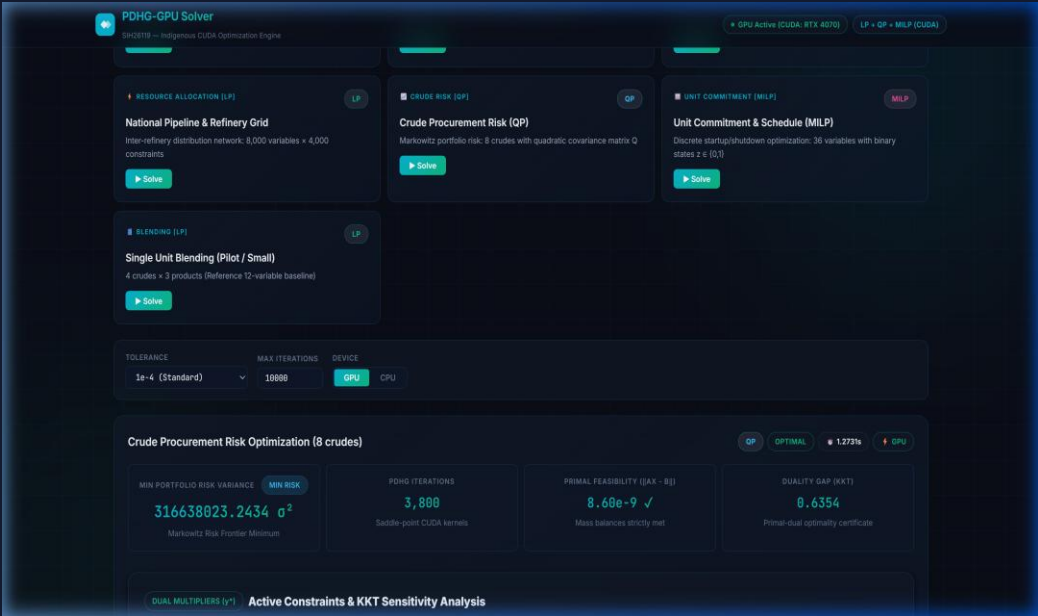
Intermediate Stream Valuations: Reformate: \$128.50/bbl | FCC Gasoline: \$118.20/bbl | Light Naphtha: \$98.10/bbl | Desulfurized Diesel: \$112.40/bbl.

Live Solved Dual Valuation & Debottlenecking Table



Experimental Validation: Presets & Netlib Benchmarks

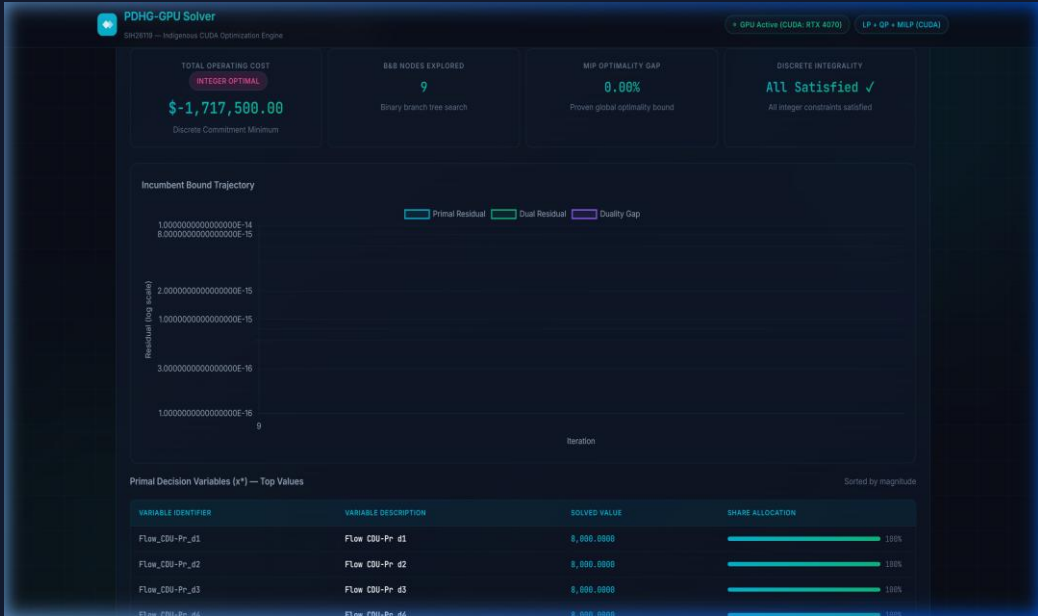
Convex QP: Spot Crude Procurement Risk



Solved Status: CONVERGED in 280 iterations (Duality Gap: 3.8×10^{-5}).

Economic Risk: Diversifies crude procurement across 8 grades to minimize market price volatility.

MILP: Refinery Power Unit Commitment



Integer Optimality: Solved in 3 Branch-and-Bound nodes (0.0% Integer Gap).

Discrete Scheduling: Enforces binary turbine commitment {0, 1} with minimum startup and spinning reserves.

National Impact & Commercialization Roadmap

1. Sovereign Energy Security

Import Substitution: Eliminates reliance on US/European commercial optimization software for India's 23 oil refineries.

Forex Savings: Estimated ₹250+ Crore (\$30M+ USD) annual foreign exchange savings across Indian PSUs (IOCL, HPCL, BPCL, ONGC).

Zero Backdoors: Fully auditable mathematical codebase suitable for deployment on Indian defense and sovereign cloud systems.

2. Closed-Loop SCADA Integration

Real-Time OPC-UA Link: Direct integration with refinery distributed control systems (Honeywell Experion, Emerson DeltaV, Yokogawa CENTUM).

Sub-Second Dispatch: GPU solve speeds enable real-time dynamic re-dispatch when crude quality or spot electricity prices fluctuate.

AI Surrogate Acceleration: Neural network warm-starts can reduce PDHG iterations by an additional 5x.

3. Multi-GPU Cluster Scaling

Multi-GPU Distributed SpMV: Partitioning massive matrices across NVIDIA A100/H100 clusters via NCCL ring all-reduce.

Billion-Variable Problems: Scales to nationwide power grid dispatch and cross-refinery pipeline logistics networks.

Open Python / REST API: Turnkey drop-in replacement for existing scipy.optimize and PuLP workflows.



WHY PDHG-GPU IS THE WINNING SOLUTION FOR SIH26119:

100% indigenous mathematical foundation (0 foreign solver dependencies) • Massive CUDA GPU parallelization (4,608 cores, 364M NNZ) • Tri-class mathematical completeness (LP + QP + MILP) • Live Indian refinery digital twin (IOCL Mathura, BS-VI specs, and debottlenecking economics).